

Library Management System - Project Documentation

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1. Overview

The **Library Management System** is a Spring Boot application designed to manage information for two primary entities: **Authors** and **Books**. It implements full CRUD operations (Create, Read, Update) and features a layered architecture consisting of Controller, Service, Repository, and Model layers.

- **GitHub URL:** <https://github.com/rikkalaudaykiranreddy/library-management>

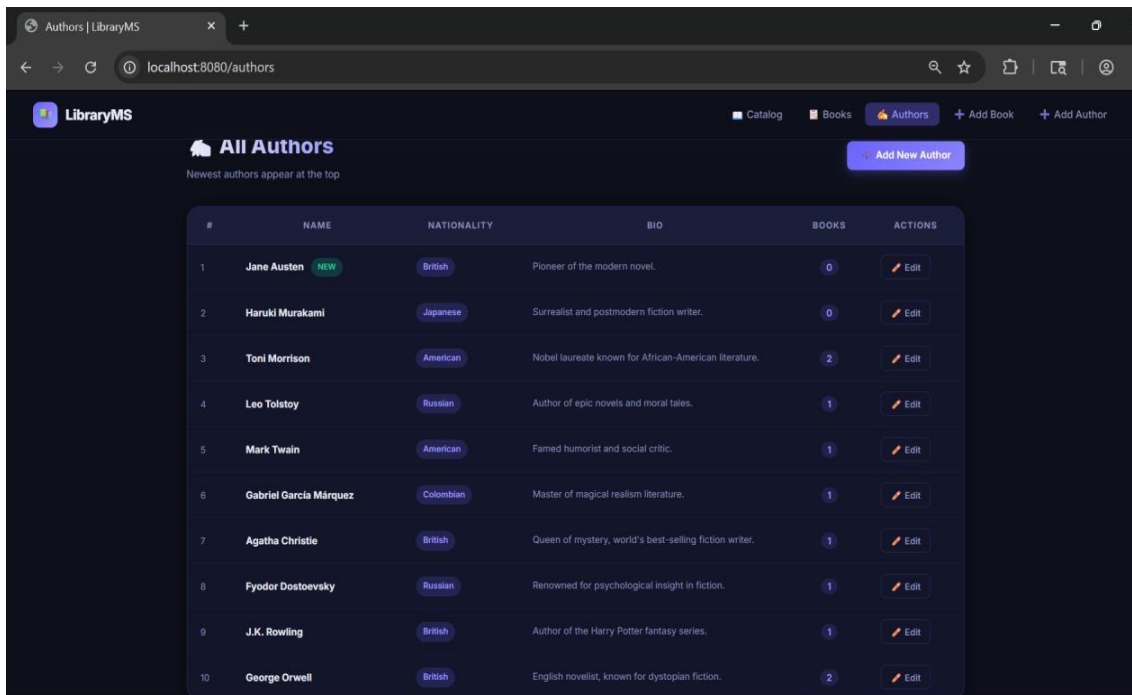
2. Entity Relationship Design

The application defines two main entities with a **One-to-Many** relationship:

- **Author:** Represents a book author. Attributes include id, name, nationality, and bio.
- **Book:** Represents a book. Attributes include id, title, genre, publicationYear, price, and isbn.

Relationship Implementation:

- An **Author** can have many **Books** (@OneToMany).
- A **Book** belongs to exactly one **Author** (@ManyToOne).
- **CascadeType.ALL** is used so that operations on authors can propagate to their books.
- **Orphan Removal** ensures that removing a book from an author's list deletes it from the database.



#	NAME	NATIONALITY	BIO	BOOKS	ACTIONS
1	Jane Austen NEW	British	Pioneer of the modern novel.	0	Edit
2	Haruki Murakami	Japanese	Surrealist and postmodern fiction writer.	0	Edit
3	Toni Morrison	American	Nobel laureate known for African-American literature.	2	Edit
4	Leo Tolstoy	Russian	Author of epic novels and moral tales.	1	Edit
5	Mark Twain	American	Famed humorist and social critic.	1	Edit
6	Gabriel García Márquez	Colombian	Master of magical realism literature.	1	Edit
7	Agatha Christie	British	Queen of mystery, world's best-selling fiction writer.	1	Edit
8	Fyodor Dostoevsky	Russian	Renowned for psychological insight in fiction.	1	Edit
9	J.K. Rowling	British	Author of the Harry Potter fantasy series.	1	Edit
10	George Orwell	British	English novelist, known for dystopian fiction.	2	Edit

#	TITLE	GENRE	YEAR	PRICE	ISBN	AUTHOR	ACTIONS
1	JASHUVA NEW	Motivational	1989	\$15.00	978-6543216789	Toni Morrison	Edit
2	Beloved	Historical	1987	\$9.49	978-1400033416	Toni Morrison	Edit
3	War and Peace	Historical	1869	\$14.99	978-1400079988	Leo Tolstoy	Edit
4	Adventures of Huckleberry Finn	Adventure	1884	\$6.99	978-0486280615	Mark Twain	Edit
5	One Hundred Years of Solitude	Magical Realism	1967	\$11.99	978-0060883287	Gabriel Garcia Márquez	Edit
6	Murder on the Orient Express	Mystery	1934	\$10.49	978-0062693662	Agatha Christie	Edit
7	Crime and Punishment	Psychological	1866	\$8.99	978-0143058144	Fyodor Dostoevsky	Edit
8	Harry Potter & Sorcerer's Stone	Fantasy	1997	\$12.99	978-0439708180	J.K. Rowling	Edit
9	Animal Farm	Political Satire	1945	\$7.49	978-0451526342	George Orwell	Edit
10	1984	Dystopian	1949	\$9.99	978-0451524935	George Orwell	Edit

3. Implementation Details

CRUD Operations

1. Create:

- Users can add new authors and books via intuitive JSP forms.
- Forms include validation (e.g., non-blank names, valid years, non-negative prices).
- BookService checks for duplicate ISBNs before saving.

+ Add New Book
Fill in the details to add a new book to the library

[-- Back to Books](#)

BOOK TITLE *
e.g. The Great Gatsby

GENRE *
e.g. Fiction

PUBLICATION YEAR *
0

PRICE (\$) *
0.0

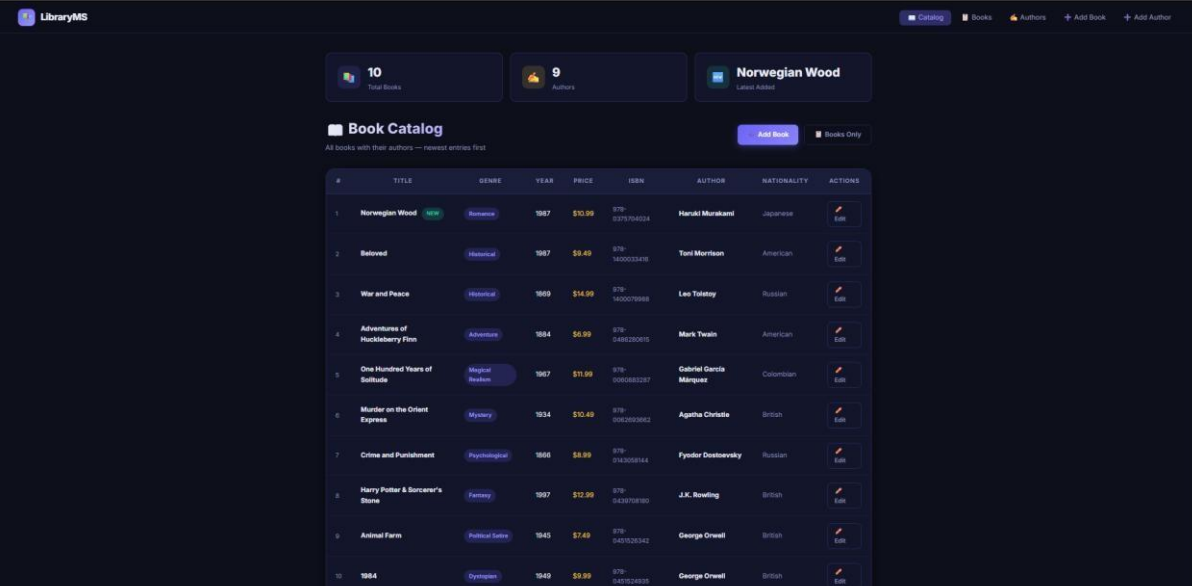
ISBN
e.g. 978-0000000000

AUTHOR *
-- Select Author --

Cancel Add Book

2 Read:

- List views for both Authors and Books.
- **Newest First Requirement:** Both lists are ordered by ID DESC, ensuring that newly added records appear at the top.
- A **Catalog View** performs an **Inner Join** between Authors and Books using a custom JPQL query in the AuthorRepository.

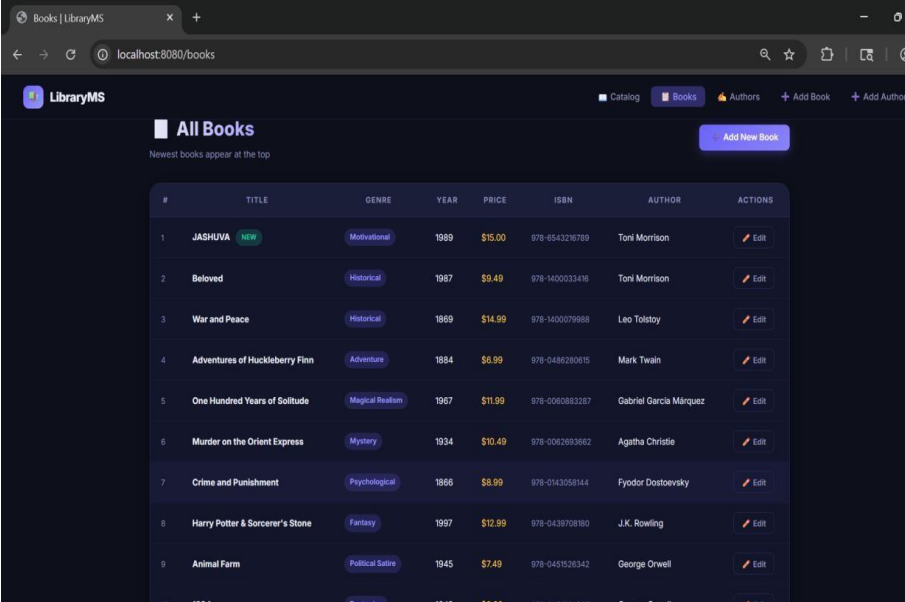


The screenshot shows the 'Book Catalog' view in the LibraryMS application. It displays a table of books with the following data:

#	TITLE	GENRE	YEAR	PRICE	ISBN	AUTHOR	NATIONALITY	ACTIONS
1	Norwegian Wood	Romance	1987	\$10.99	978-0375764224	Haruki Murakami	Japanese	Edit
2	Beloved	Historical	1987	\$9.49	978-1400033439	Toni Morrison	American	Edit
3	War and Peace	Historical	1869	\$14.99	978-1400079888	Leo Tolstoy	Russian	Edit
4	Adventures of Huckleberry Finn	Adventure	1884	\$6.99	978-0486280615	Mark Twain	American	Edit
5	One Hundred Years of Solitude	Magical Realism	1967	\$11.99	978-0060863287	Gabriel Garcia Márquez	Colombian	Edit
6	Murder on the Orient Express	Mystery	1934	\$10.49	978-0062693662	Agatha Christie	British	Edit
7	Crime and Punishment	Psychological	1866	\$8.99	978-0143058144	Fyodor Dostoevsky	Russian	Edit
8	Harry Potter & Sorcerer's Stone	Fantasy	1997	\$12.99	978-0439708180	J.K. Rowling	British	Edit
9	Animal Farm	Political Satire	1945	\$7.49	978-0431026342	George Orwell	British	Edit
10	1984	Dystopian	1949	\$9.99	978-0451524935	George Orwell	British	Edit

3 Update:

- Users can edit existing records.
- The Update logic ensures data integrity (e.g., not allowing duplicate ISBNs even during updates, excluding the record itself).



The screenshot shows the 'All Books' view in the LibraryMS application. It displays a table of books with the following data:

#	TITLE	GENRE	YEAR	PRICE	ISBN	AUTHOR	ACTIONS
1	JASHUVA	Motivational	1989	\$15.00	978-0543216789	Toni Morrison	Edit
2	Beloved	Historical	1987	\$9.49	978-1400033436	Toni Morrison	Edit
3	War and Peace	Historical	1869	\$14.99	978-1400079888	Leo Tolstoy	Edit
4	Adventures of Huckleberry Finn	Adventure	1884	\$6.99	978-0486280615	Mark Twain	Edit
5	One Hundred Years of Solitude	Magical Realism	1967	\$11.99	978-0060863287	Gabriel Garcia Márquez	Edit
6	Murder on the Orient Express	Mystery	1934	\$10.49	978-0062693662	Agatha Christie	Edit
7	Crime and Punishment	Psychological	1866	\$8.99	978-0143058144	Fyodor Dostoevsky	Edit
8	Harry Potter & Sorcerer's Stone	Fantasy	1997	\$12.99	978-0439708180	J.K. Rowling	Edit
9	Animal Farm	Political Satire	1945	\$7.49	978-0451526342	George Orwell	Edit
10	1984	Dystopian	1949	\$9.99	978-0451524935	George Orwell	Edit

Layered Architecture

- **Model:** JPA Entities (Author.java, Book.java) and DTOs (BookAuthorDTO.java).
- **Repository:** Spring Data JPA interfaces (AuthorRepository.java, BookRepository.java) with custom JPQL queries.
- **Service:** Business logic (AuthService.java, BookService.java) with @Transactional support.
- **Controller:** MVC Controllers (AuthorController.java, BookController.java) handling HTTP requests and view binding.
- **View:** JSP pages with CSS styling for a premium look.

Custom Query (Inner Join)

The AuthorRepository includes a custom query to fetch combined data:

java

```
@Query("SELECT new com.library.dto.BookAuthorDTO(" +  
    "b.id, b.title, b.genre, b.publicationYear, b.price, b.isbn, " +  
    "a.id, a.name, a.nationality) " +  
    "FROM Book b INNER JOIN b.author a " +  
    "ORDER BY b.id DESC")
```

```
List<BookAuthorDTO> findAllBooksWithAuthors();
```

4. Testing and Validation

- **Unit Testing:** 29 tests covering Repository and Service layers using JUnit 5 and Mockito.
- **Validation:** Jakarta Validation annotations (e.g., @NotBlank, @Min, @DecimalMin) ensure data quality.
- **Exception Handling:** A custom LibraryException wraps data integrity violations and provides user-friendly error messages via flash attributes.

5. Challenges Faced

- **Newly Added at Top:** Implementing this required custom sorting in the service layer for findAll() results and ORDER BY clauses in custom JPQL queries.
- **Duplicate ISBN Detection:** Specifically during updates, the logic needed to exclude the current record's ID to avoid false positives.
- **JSP Immutability:** Encountered an UnsupportedOperationException when trying to sort an immutable list returned by a mock in unit tests. Resolved by creating a mutable ArrayList copy in the service layer.

6. Visual Presentation

- The application uses a **Modern Dark Theme** with glassmorphism effects.
- **Badges:** New entries are highlighted with a "NEW" badge.
- **Stats:** The catalog view includes real-time stats on total books and unique authors.